



Market-Oriented Data Allocation and New Quality Productive Forces in China: Dynamic Configurational Pathways for Urban and Regional Development



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Received: 10-13-2025

Revised: 12-10-2025

Accepted: 12-17-2025

Citation: C. Wang and M. Y. Lu, “Market-oriented data allocation and new quality productive forces in China: Dynamic configurational pathways for urban and regional development,” *J. Urban Dev. Manag.*, vol. 4, no. 4, pp. 291–314, 2025. <https://doi.org/10.56578/judm040405>.



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Abstract: Market-oriented data allocation is becoming an important institutional foundation for digital transformation, innovation-led growth, and coordinated urban and regional development. However, its contribution to new quality productive forces depends on the joint operation of data-market infrastructure, government data openness, digital capacity, enterprise participation, institutional conditions, and policy support rather than on any single factor. This study investigates how these conditions combine to produce high levels of regional new quality productive forces and how the resulting pathways vary across time and regions. Provincial panel data covering 31 provincial-level regions in China from 2018 to 2023 were analyzed using dynamic qualitative comparative analysis (QCA). The results showed that none of the six antecedent conditions constituted an individually necessary condition. Six sufficient configurations were identified and grouped into three models: the Scale–Environment Model, the Level–Scale–Environment Model, and the Level–Scale Model. The overall solution achieved a consistency of 0.929 and coverage of 0.792. The configuration-level analysis further showed that the principal pathways remained broadly stable over time, although their relative importance changed during the study period. Regional comparison revealed that most configurations showed similar applicability across regions, whereas one Level–Scale configuration displayed significant regional heterogeneity and was more strongly associated with western China. These findings indicate that the development of new quality productive forces requires locally adapted combinations of market institutions, data openness, digital infrastructure, and enterprise participation. The study provides a configurational explanation of how data-factor markets support urban and regional development and offers practical guidance for coordinating digital governance, innovation policy, and data resource allocation across regions with different development conditions.

Keywords: Market-oriented data allocation; New quality productive forces; Urban and regional development; Digital governance; Institutional environment; Dynamic qualitative comparative analysis

1 Introduction

The concept of “new quality productive forces” emerged in discussions following the September 2023 inspection tour of Heilongjiang Province and has been associated with efforts to promote high-quality economic and social development. New quality productive forces describe an advanced form of productivity emerging in the digital economy through technological breakthroughs, the reorganization of production factors, and the transformation and upgrading of industrial systems. The subsequent call to “integrate scientific and technological innovation resources, lead the development of strategic emerging industries and future industries, and accelerate the formation of new quality productive forces” further emphasized the central role of technological innovation in this development agenda. Unlike conventional growth models that rely heavily on capital accumulation and resource consumption, new quality productive forces depend on new technologies, new forms of labor, new production instruments, and more efficient combinations of productive resources. Data have become particularly important in this process. As a production input, data differ from conventional factors because they are largely non-rival and can be reused across firms, sectors, and applications. Their economic value therefore depends not only on data generation but also on the institutional arrangements governing access, circulation, and reuse [1]. However, the economic value of data cannot be realized through accumulation alone. It also requires market institutions capable of clarifying property

rights, establishing transaction rules, protecting participating entities, reducing barriers to circulation, and supporting the lawful exchange and reuse of data. Effective data governance further requires clearly defined decision rights, accountability structures, access arrangements, and control mechanisms. These institutional arrangements can reduce uncertainty in data sharing and support the responsible reuse of data across organizational boundaries [2]. Understanding how market-oriented data allocation contributes to this process is therefore important for both economic policy and development management.

Market-oriented data allocation is more than a reform of factor-allocation mechanisms. It is also an institutional arrangement with direct consequences for digital governance, industrial transformation, and coordinated urban and regional development. From a regional economic perspective, the circulation of data across administrative and sectoral boundaries can support the movement and recombination of capital, technology, talent, and knowledge, thereby reshaping regional growth drivers and industrial spatial structures. These effects are particularly relevant to urban regions, where digital industries, public services, infrastructure networks, and innovation resources are highly concentrated but are often governed through fragmented administrative and institutional arrangements. From the perspective of digital transformation, clearer property rights, standardized transactions, and more transparent pricing mechanisms can reduce the institutional costs associated with acquiring and using data. These arrangements create more favorable conditions for technological application, business-model restructuring, organizational change, and the coordinated development of industrial digitalization and digital industries. From a governance perspective, market-oriented allocation can connect the controlled opening of public data with the orderly circulation of social and commercial data. This connection can improve the informational basis of public decision-making, interdepartmental coordination, urban service provision, and the management of development resources. It also supports a transition from governance based largely on administrative experience to governance informed by timely and reusable data. Market-oriented data allocation should therefore be understood not only as an economic mechanism for cultivating new quality productive forces, but also as part of the institutional infrastructure through which cities and regions manage digital transformation and pursue differentiated development.

Existing studies in China have examined the relationship between data allocation and new quality productive forces at the enterprise, industrial, and regional levels. At the firm level, big data analytics capabilities can support organizational performance by strengthening the dynamic capabilities through which firms integrate, reconfigure, and apply digital resources [3]. At the industry level, digital technologies alter how innovation activities are organized by enabling distributed participation, recombination of digital resources, and interaction across organizational and sectoral boundaries [4]. At the regional level, the economic effects of digital technologies depend on complementary infrastructure, human capital, industrial structure, and institutional capacity. Because these complementary conditions are unevenly distributed, the productivity effects of digitalization may vary considerably across regions [5]. This spatial heterogeneity suggests that the pathways through which the market-oriented allocation of data elements contributes to regional new quality productive forces are unlikely to be uniform. Despite these advances, most studies treat the relevant conditions as independent explanatory variables and estimate their average net effects. Such an approach is less suited to situations in which institutional quality, digital infrastructure, government data openness, enterprise participation, and policy support operate jointly and can substitute for or complement one another. Existing research also provides limited evidence on whether the combinations associated with high levels of new quality productive forces remain stable over time or vary across regions with different economic and institutional conditions. These limitations are important from an urban and regional development perspective because a policy arrangement that works in a highly digitalized coastal region may not be equally suitable for an inland region with a different market structure, infrastructure base, or governance capacity. A dynamic configurational approach is therefore required to identify multiple development pathways, examine their temporal stability, and determine the extent to which their explanatory power depends on regional context.

To address these issues, this study draws on transaction cost theory, resource orchestration theory, and institutional theory and applies dynamic qualitative comparative analysis to panel data from 31 provincial-level regions in China between 2018 and 2023. The analysis examines how six conditions—the number of big data exchanges, government data openness, informatization, the share of e-commerce enterprises, the marketization environment, and government policy support—combine to produce high levels of regional new quality productive forces. It also evaluates the temporal stability of the identified configurations and investigates whether their explanatory power differs across regions. The study makes two principal contributions. First, it moves beyond the assumption that data-market conditions operate independently by developing a configurational framework that explains how marketization level, marketization scale, and marketization environment work together. This framework identifies alternative routes through which regions with different institutional and economic foundations can convert data resources into productive capacity. It also connects the development of data-factor markets with broader questions of digital governance, urban–regional coordination, and place-sensitive resource allocation. Second, the use of dynamic qualitative comparative analysis (QCA) incorporates temporal and spatial variation into the configurational analysis. Rather than treating the identified pathways as static or universally applicable, the study examines their

stability over the observation period and their applicability across eastern, central, and western China. In doing so, it extends the empirical use of dynamic configurational analysis in regional development research and provides a more context-sensitive basis for designing data-allocation, digital-governance, and innovation policies.

2 Literature Review

2.1 Market-Oriented Allocation of Data Elements

The rapid expansion of the digital economy has established data as an increasingly important input into production, innovation, and economic decision-making. Unlike conventional inputs that are depleted through use, data can be repeatedly reused, processed, and applied across different contexts without being exhausted. Their economic value therefore depends not only on their initial collection but also on their effective utilization and combination with complementary productive resources [6]. Data do not create economic value merely through their existence or accumulation. Their productive value emerges when they are transformed into useful information through collection, processing, and analysis, and when they are combined with complementary technological, human, and organizational capabilities. Because digital resources can be recombined across actors, processes, and applications, they can support new forms of innovation and reshape how productive resources are organized and deployed [7].

Market-oriented allocation establishes institutional and transactional arrangements for data circulation. Clearer access rights, standardized transaction procedures, functional data platforms, and transparent pricing mechanisms can reduce frictions in data exchange and make data more accessible to public and private actors [2]. Recent evidence indicates that data element marketization can improve urban green total factor energy efficiency by stimulating technological innovation and strengthening environmental governance [8]. This relationship is particularly relevant to cities and urban regions, where economic activity, digital infrastructure, and large volumes of public and commercial data are spatially concentrated.

The distinctive economic properties of data provide a further basis for understanding these outcomes. Data can be used repeatedly and simultaneously across different economic activities, allowing the same information resources to contribute to multiple production, innovation, governance, and service processes [9]. This capacity for multiple use distinguishes data from conventional rival inputs and creates the potential for increasing returns when access and reuse are effectively organized. Data can also serve as an input into research, knowledge accumulation, and innovation, linking data generation and application to productivity and economic growth [10]. At the same time, these effects are not automatic, because ownership arrangements, privacy constraints, access rules, and institutional incentives influence how extensively and efficiently data are used. The developmental contribution of data therefore depends not only on their availability but also on the institutional environment through which they are accessed, exchanged, governed, and combined with other productive resources.

2.2 Research on New Quality Productive Forces

New quality productive forces have recently emerged as an important concept in China's policy discourse on productivity transformation and high-quality development. Officially, the concept refers to an innovation-led form of advanced productivity that departs from development based primarily on extensive factor inputs and resource consumption [11]. Its central features include technological innovation, industrial upgrading, improved factor allocation, digital transformation, and more sustainable forms of production. Because the concept is recent and specific to the Chinese development context, a settled definition has not yet emerged in the international literature. Nevertheless, its principal components are closely connected to established research on digital transformation, technological change, and the reorganization of production. Digital transformation changes how organizations combine resources, coordinate activities, and create value [12], while technological innovation and automation alter the allocation of tasks between labor and technology and generate new skill requirements [13]. From this perspective, new quality productive forces are treated in this study as a multidimensional regional productive capacity shaped by technological upgrading, digital integration, and environmentally sustainable development.

The digital economy extends this discussion by drawing attention to the complementary conditions required for productivity gains to emerge. Digital technologies can reduce information and transaction costs and change how economic activities are organized [14], but their effects depend on infrastructure, skills, organizational adjustment, and access to markets. Evidence on broadband expansion shows that digital connectivity can influence employment and skill demand, although the magnitude of these effects varies across locations and sectors [15]. Productivity gains from general-purpose digital technologies may also appear only after complementary investments have been made in human capital, organizational processes, and intangible assets [16]. Environmental outcomes are similarly conditional: digitalization can support monitoring and resource efficiency, but additional equipment, infrastructure, and energy demand may offset part of these gains [17]. These findings suggest that the development of new quality productive forces is unlikely to follow a uniform regional pathway. Differences in infrastructure, skills, industrial structure, market access, and institutional capacity may cause similar policy initiatives to produce different outcomes across provinces and urban regions.

The measurement of new quality productive forces remains unsettled because the concept combines productivity, innovation, digitalization, and environmental performance. Total factor productivity captures changes in aggregate production efficiency, but it does not directly distinguish the technological, digital, and environmental dimensions emphasized in the present conceptualization. A composite index is therefore used in this study to represent the outcome as a multidimensional regional construct. Composite-index design requires an explicit conceptual framework, transparent indicator selection, appropriate normalization and weighting, and careful interpretation of the resulting scores [18]. Particular caution is required because assigned weights do not necessarily correspond to the effective importance of individual indicators within the final index [19]. Accordingly, regional new quality productive forces are operationalized here through three dimensions: technological productivity, green productivity, and digital productivity. These dimensions are treated as the measurement framework adopted in this study rather than as a universally established classification. Technological productivity represents innovation inputs and outputs; green productivity captures resource use and environmental performance; and digital productivity reflects digital infrastructure, digital industries, and the application of digital technologies.

2.3 Mechanisms Linking Market-Oriented Data Allocation to New Quality Productive Forces

Data occupy several positions in contemporary production systems. They serve as inputs for analysis, forecasting, coordination, and automated decision-making, while their collection, classification, processing, and commercialization have also become economically significant activities. Firm-level evidence has linked the use of big-data analytics to innovation outcomes, although access to large volumes of data does not by itself ensure that firms can convert them into economically useful knowledge [20]. This combination of productive use and continuing accumulation has led scholars to conceptualize data as a form of capital whose value is created through extraction, aggregation, and application [21]. A different perspective treats user-generated data as a form of labor and considers how the economic value derived from individual data contributions should be distributed [22]. Endogenous-growth research offers a further explanation by regarding data as a reusable input into research, knowledge accumulation, and innovation [10]. Despite their different theoretical foundations, these perspectives point to a common conclusion: the availability of data alone does not generate productivity. Its productive value depends on the institutional and organizational arrangements through which data are collected, accessed, processed, governed, and incorporated into production and innovation.

As data markets have developed, greater attention has been paid to the institutional arrangements through which external data are exchanged and incorporated into productive activities. Data marketplaces provide structured settings in which data providers, intermediaries, and potential users interact. Depending on their business models, these marketplaces may facilitate data discovery, access, licensing, pricing, and exchange [23]. However, data marketplaces differ considerably in their participants, revenue models, pricing mechanisms, access conditions, and governance structures. Their establishment does not in itself guarantee effective data circulation. The use and exchange of data also depend on data quality, interoperability, clearly defined usage rights, privacy protection, security safeguards, and the participation of credible providers and users. These requirements are particularly important in urban and regional contexts, where public, industrial, and commercial data are distributed across multiple organizations and administrative jurisdictions. Effective cross-organizational data use therefore requires governance arrangements that clarify decision rights, responsibilities, standards, and accountability [2].

At the industrial and enterprise levels, data allocation can influence productive capacity through digital transformation, innovation, information disclosure, and supply-chain coordination. Evidence from China indicates that public data openness is associated with stronger enterprise new quality productive forces through improvements in information-disclosure quality, innovation quality, and supply-chain efficiency [24]. However, access to external data does not automatically translate into productive outcomes. Firms require analytical, managerial, and organizational capabilities to interpret data, combine them with internal resources, and translate the resulting information into operational and strategic decisions. From a dynamic-capability perspective, data-related capabilities can support the identification of technological and market changes and the reconfiguration of organizational resources and processes [3]. The productive contribution of market-oriented data allocation therefore depends on the interaction of external data access, digital infrastructure, organizational capacity, market participation, and institutional governance.

Existing studies thus identify several channels through which data allocation may contribute to productive transformation, including data access, digital transformation, innovation, supply-chain coordination, and organizational reconfiguration. However, these channels are generally examined separately and are commonly represented as net or linear effects. Less is known about how institutional, technological, market, and policy conditions operate jointly across different regional settings. This limitation is important because provinces and urban regions differ substantially in government-data openness, digital infrastructure, market participation, data-exchange development, institutional quality, and policy support. It is therefore unlikely that a single condition or uniform policy model will be sufficient across all regions. A configurational perspective is required to identify the alternative combinations of conditions associated with high regional new quality productive forces.

2.4 Literature Review Summary

Existing research provides a substantial foundation for understanding the economic and developmental consequences of market-oriented data allocation. Studies have examined its influence on resource-allocation efficiency, green transformation, technological innovation, enterprise digitalization, and spatial spillovers. Research on new quality productive forces has likewise progressed from single indicators toward multidimensional evaluation systems that capture technological, green, and digital forms of productivity. Together, these strands of literature explain several channels through which data resources may be converted into regional productive capacity and provide an empirical basis for measuring both data-market development and new quality productive forces.

Several questions nevertheless remain unresolved. First, most empirical studies rely on econometric models designed to estimate the independent net effect of individual variables. Such models are useful for identifying average relationships but are less able to explain causal complexity when several conditions must be present together or when one condition can compensate for the absence of another. The configurations through which data-market infrastructure, government data openness, informatization, enterprise participation, market institutions, and policy support jointly contribute to new quality productive forces therefore remain insufficiently examined.

Second, current mechanism studies generally focus on individual mediation or moderation relationships. This makes it difficult to determine whether different regions can reach comparable outcomes through alternative combinations of conditions. It also leaves the interdependence among marketization level, marketization scale, and marketization environment underdeveloped. A region with limited data-trading infrastructure, for example, may rely more heavily on enterprise participation and a well-functioning market environment, whereas a region with stronger digital infrastructure may follow a different development path. Identifying these alternative pathways is important for urban and regional policy because development conditions, governance capacity, and institutional maturity vary substantially across locations.

Third, much of the existing evidence is static. Even where regional differences are considered, limited attention is given to whether the identified relationships remain stable over time or whether their explanatory power changes across spatial contexts. This is a significant limitation because data markets, digital infrastructure, policy systems, and enterprise participation evolve rapidly. A configuration that supports regional development in one period may weaken, strengthen, or be replaced as institutional and technological conditions change.

To address these gaps, this study applies dynamic qualitative comparative analysis from a configurational perspective. It examines how multiple concurrent conditions form alternative sufficient pathways to high levels of regional new quality productive forces and evaluates the temporal stability and regional applicability of those pathways. This design makes it possible to examine causal conjunction, equifinality, and asymmetric relationships that cannot be fully represented by conventional net-effect models. It also connects data-market development with the wider concerns of digital governance and urban and regional development. The resulting analysis provides a more context-sensitive basis for understanding how regions with different institutional, market, and digital foundations can design data-allocation policies suited to their own development conditions.

3 Theoretical and Analytical Framework

Transaction-cost economics explains why market exchange involves costs beyond the direct costs of producing goods or services. These costs arise from searching for information and trading partners, negotiating agreements, monitoring performance, and enforcing contractual obligations [25]. They are particularly relevant to data transactions because the quality and prospective value of data are often difficult to assess before use. Uncertainty may also arise over access rights, permitted uses, transferability, pricing, privacy, security, and liability. The non-rival and reproducible nature of data does not eliminate these problems; instead, it may create additional difficulties in defining rights and monitoring subsequent use.

Data exchanges and government open-data platforms can be understood as institutional responses to some of these transaction problems. Data exchanges act as intermediaries between providers and potential users and may offer product descriptions, licensing arrangements, transaction procedures, pricing services, and compliance mechanisms [23]. Government open-data platforms provide access to public-sector information and may reduce the search and acquisition costs faced by firms, research institutions, civic organizations, and other users. However, neither type of platform necessarily produces efficient data circulation. Its effectiveness depends on data quality, interoperability, contractual clarity, privacy protection, user participation, and enforcement capacity. In urban and regional economies, where relevant data are distributed across public agencies, infrastructure operators, industries, and service providers, these arrangements may reduce fragmentation and create more reliable channels for data access and exchange. From a transaction-cost perspective, the number of operational data exchanges and the degree of government data openness therefore reflect different aspects of the institutional basis for data circulation.

Resource orchestration theory shifts attention from the possession of resources to the processes through which resources are structured, bundled into capabilities, and deployed in response to changing conditions [26]. This distinction is important because the availability of data does not, by itself, generate innovation or productivity. Data

must be obtained from appropriate sources, processed and integrated, combined with complementary resources, and incorporated into production, governance, or decision-making activities. Market-oriented data allocation consequently involves more than the release or activation of previously unused data. It also concerns the capacity of market actors to combine data with technology, skills, organizational knowledge, and application opportunities.

Resource orchestration was originally developed mainly at the firm level. In this study, its regional application refers to the coordinated mobilization of digital infrastructure, human capital, market participants, and institutional resources rather than to the decisions of a single firm. Within this regional interpretation, the share of e-commerce enterprises and the level of informatization represent two conditions associated with the scale and depth of productive data use. E-commerce enterprises are substantial producers and users of transaction, consumer, logistics, and platform data. Their presence indicates the availability of market actors and commercial settings in which data are generated and repeatedly applied. Regional informatization reflects the technical and human-capital foundations required to store, transmit, process, and integrate data across firms and sectors. These conditions may be complementary: a large number of digitally active firms may generate limited productive value when infrastructure and technical skills are insufficient, while advanced infrastructure may remain underused when market participation and application scenarios are limited. From a resource-orchestration perspective, their interaction influences whether fragmented data resources can be converted into capabilities that support regional production, urban innovation, and digitally mediated services.

Institutional theory directs attention to the formal rules, public policies, shared expectations, and enforcement structures that shape economic behavior. Market-oriented data allocation does not develop as a purely spontaneous market process. It is embedded in arrangements that determine who may control and access data, how transactions are conducted, which forms of reuse are permitted, and how risks and responsibilities are distributed. The regional marketization environment represents the broader institutional setting in which these activities take place. It includes the relationship between government and the market, the development of the non-state sector, the maturity of product and factor markets, and the strength of market intermediaries and legal institutions.

Government policy support represents a more direct form of institutional intervention. Local governments participate in the development of data markets through public-data policies, data-rights arrangements, transaction rules, security requirements, infrastructure investment, and regulatory oversight. These actions influence market participants' expectations concerning the costs, returns, risks, and legitimacy of data-related activities. They are also central to urban digital governance because local authorities must reconcile economic use and data circulation with public accountability, privacy, information security, and responsible management of public resources. The marketization environment and policy support are therefore treated as related but distinct conditions: the former reflects the general quality of regional market institutions, whereas the latter captures the specific policy attention directed toward data circulation and use.

The three theoretical perspectives explain different but interdependent components of market-oriented data allocation. Transaction-cost economics explains how exchange institutions and public-data access may alter the costs and uncertainty associated with obtaining and using data. Resource orchestration theory explains why data generate productive outcomes only when they are combined with infrastructure, skills, market actors, and application opportunities. Institutional theory explains how market development, policy support, legal arrangements, and public governance shape the legitimacy and feasibility of these activities. None of the three perspectives is sufficient on its own. Data may be accessible but remain underused when technical and organizational capacity is weak. Firms may possess substantial digital capabilities but encounter high transaction costs, uncertain access rights, or limited external data availability. Similarly, policy support may produce limited results when market participation, institutional credibility, or practical application scenarios are absent.

Market-oriented data allocation is therefore understood as a regional system in which transactional arrangements, productive capabilities, and institutional conditions operate jointly. This interpretation is consistent with the argument that data do not generate new quality productive forces through availability alone. Their contribution depends on whether regions can establish workable channels for data circulation, mobilize complementary resources, and provide an institutional environment that permits data to be used securely and productively.

Based on this theoretical integration, the study constructs a three-dimensional analytical framework comprising marketization level, marketization scale, and marketization environment, as illustrated in Figure 1. Marketization level is represented by the number of big data exchanges and the degree of government data openness. These indicators capture two channels through which data become available: organized market exchange and public-sector data provision. The number of exchanges reflects the presence of formal venues for data transactions, whereas government data openness reflects the accessibility and usability of public data. Neither indicator alone measures the overall maturity or performance of the data market, but together they represent the institutional basis for data access and circulation.

Marketization scale is represented by the share of e-commerce enterprises and the level of informatization. The share of e-commerce enterprises indicates the presence of digitally active market participants and commercially established data-application scenarios. Informatization reflects the regional foundations for data transmission,

processing, storage, and integration. Together, these conditions represent the breadth of market participation and the technical capacity through which data can be incorporated into economic activity. Marketization environment is represented by the regional marketization environment index and government policy support. The marketization environment index captures the general quality of regional market institutions, whereas government policy support represents the attention given by local authorities to data-market construction and governance. These conditions influence whether market actors regard data transactions as feasible, predictable, and institutionally legitimate.

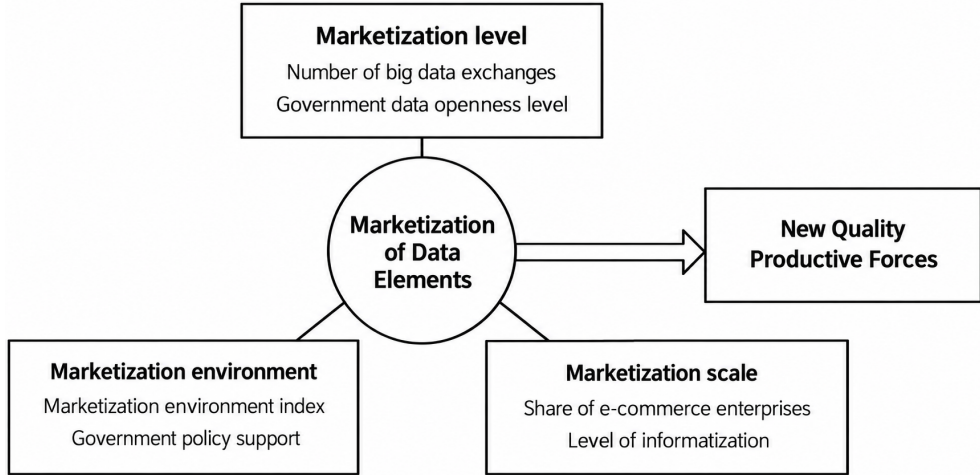


Figure 1. Conceptual framework for the configurational analysis

The three dimensions are analytically distinct but functionally interdependent. A favorable institutional environment may support data circulation, but its influence may remain limited without adequate digital capacity or active market participants. Conversely, a high level of informatization may not translate into new quality productive forces when data access is restricted, transaction institutions are underdeveloped, or firms lack viable application settings. Formal data exchanges may also have limited practical importance when trading activity, data quality, or user participation remains low.

Different regions may therefore reach similarly high levels of new quality productive forces through different combinations of conditions. This expectation is consistent with the substantial economic and institutional diversity of Chinese provinces and with the place-sensitive nature of urban and regional development. The framework does not assume that the six conditions operate independently or that their effects are linearly additive. Instead, regional new quality productive forces are treated as the outcome of alternative configurational pathways formed through the interaction of marketization level, marketization scale, and marketization environment.

4 Research Design

4.1 Research Method

Qualitative comparative analysis adopts a set-theoretic and configurational approach to causal explanation. Rather than estimating the independent net effect of each explanatory variable, QCA identifies combinations of conditions that are sufficient or necessary for the presence or absence of an outcome. It is particularly appropriate for investigating conjunctural causation, causal asymmetry, and equifinality—that is, situations in which conditions operate interdependently, the causes of an outcome differ from those of its absence, and more than one configuration can produce the same outcome [27]. These methodological properties correspond closely to the theoretical argument developed in this study. High regional new quality productive forces are unlikely to arise from any single aspect of data marketization. Instead, data-market infrastructure, government data openness, informatization, enterprise participation, market institutions, and policy support may complement or substitute for one another under different regional conditions.

Conventional QCA applications are generally cross-sectional and comparatively insensitive to temporal change. They can identify configurations associated with an outcome at a particular point or over a pooled observation period, but they cannot by themselves establish whether those configurations remain stable as institutional, technological, and market conditions evolve [28]. This limitation is consequential in the present study because China’s data-factor markets changed substantially between 2018 and 2023, while provincial-level regions differed in their digital infrastructure, industrial structure, market maturity, and governance capacity. Treating all province–year observations as temporally homogeneous could therefore obscure both changes in configurational relationships over time and persistent differences among regions.

To address this limitation, this study adopted the panel-data set-theoretic approach proposed by Garcia-Castro and Ariño [29], commonly referred to in subsequent applications as dynamic QCA. The approach extends conventional QCA by distinguishing pooled, between, and within measures of consistency and coverage. Pooled consistency and coverage describe the set relationship across all province–year observations without separating temporal and regional variation. Between consistency and coverage are calculated for each year across provinces; variations in these measures indicate whether a necessity or sufficiency relationship changes across observation periods. Within consistency and coverage are calculated for each province across years; variations in these measures reveal whether the relationship is affected by province-specific characteristics. Adjusted consistency distances were further used to evaluate the magnitude of deviations from the corresponding pooled relationship and, consequently, to identify possible temporal or regional heterogeneity.

Dynamic QCA was therefore used to identify alternative configurations of marketization-level, marketization-scale, and marketization-environment conditions associated with high regional new quality productive forces. It was also used to determine whether the identified configurations remained stable over the 2018–2023 period and whether their explanatory performance varied across provincial-level regions. This research design is consistent with the study’s objective of identifying place-sensitive pathways relevant to digital governance and urban and regional development, rather than assuming that a single model of market-oriented data allocation is universally applicable.

4.2 Sample Selection and Data Sources

The empirical sample comprised 31 provincial-level administrative regions in mainland China observed annually from 2018 to 2023. Hong Kong, Macao, and Taiwan were excluded because the statistical definitions and data-reporting systems applicable to these regions are not directly comparable with those used for mainland provincial-level units. The final dataset contained 186 province–year observations and retained both temporal variation within provinces and cross-sectional variation among provinces. The six-year period also covered an important stage in the development of China’s data-factor market, during which provincial regions differed substantially in digital infrastructure, industrial development, market institutions, and policy capacity.

Data for the outcome and antecedent conditions were obtained primarily from the 2019–2024 editions of the China Statistical Yearbook [30], the China Statistical Yearbook on Environment [31], and the China Energy Statistical Yearbook [32], together with the Wind Economic Database [33] and official provincial government websites. The statistical yearbooks and Wind database supplied the principal socioeconomic, industrial, technological, energy, and environmental indicators. Data that were not available from standardized statistical databases were manually collected from provincial government documents, development plans, government work reports, policy documents, and official announcements. These sources were used primarily to identify the establishment of data exchanges and to measure provincial policy support for the market-oriented allocation of data elements.

During data compilation, isolated missing values were identified in several annual province-level indicator series. Linear interpolation was applied only to internal gaps for which observations were available immediately before and after the missing year for the same province and indicator. No extrapolation was performed at the beginning or end of a provincial time series, and interpolation was not used to reconstruct an entirely unavailable indicator series. This procedure preserved the original sample structure without generating observations outside the available temporal range. Following data verification and the treatment of the internal gaps, a balanced panel of 186 province–year observations was retained for the dynamic QCA.

4.3 Variable Measurement

4.3.1 Outcome variable

Regional new quality productive forces were measured through three dimensions: scientific and technological productivity, green productivity, and digital productivity [34]. Scientific and technological productivity captures regional innovation inputs, technological production, and the conversion of research activity into economically usable output. Green productivity reflects resource and energy efficiency together with the environmental consequences of production. Digital productivity represents the development of digital industries, telecommunications, software services, network infrastructure, and digital commercial activity.

These dimensions correspond to three central features of new quality productive forces: innovation-driven development, environmentally sustainable production, and the integration of digital technologies with economic activity. They are also relevant to urban and regional development because technological resources, environmental pressures, digital infrastructure, and industrial activity are unevenly distributed across provincial regions and shape the development capacity of the cities located within them.

The Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) method, with entropy-based weighting, was applied to province–year indicator matrix. All positive and negative indicators were first normalized according to their directional attributes. Entropy weights were then calculated from the variation in each indicator across the complete sample, ensuring that a common weighting structure was maintained across provinces and years.

TOPSIS was subsequently used to measure the relative distance of each province–year observation from the positive and negative ideal solutions. The resulting closeness coefficient served as the outcome variable representing the level of regional new quality productive forces. The indicator definitions, measurement units, data directions, and data sources are reported in Table 1.

Table 1. Evaluation indicator system for new quality productive forces

First-Level	Second-Level	Third-Level	Definition	Unit	Attribute
Technology productivity	Innovation productivity	Innovation R&D	Number of domestic patents granted	No.	+
		Innovation industry	High-tech industry business revenue	10,000 yuan	+
		Innovation products	Innovation funds of industrial enterprises above designated size	10,000 yuan	+
	Technological productivity	Technical efficiency	Labor productivity of industrial enterprises above designated size	%	+
		Technology R&D	Full-time equivalent of R&D personnel in industrial enterprises above designated size	Hour	+
		Technology production	Raw robot installation density	%	+
Green productivity	Resource-saving productivity	Energy intensity	Energy consumption per unit of GDP	%	—
		Energy structure	Fossil energy consumption per unit of GDP	%	—
		Water intensity	Industrial water use per unit of GDP	%	—
	Environmentally friendly productivity	Waste utilization	Comprehensive utilization rate of industrial solid waste; comprehensive utilization of industrial solid waste / generation	%	+
		Wastewater discharge	Industrial wastewater discharge per unit of GDP	%	—
		Exhaust gas emission	Industrial SO ₂ emissions per unit of GDP	%	—
Digital productivity	Digital industry productivity	Electronic information manufacturing	Integrated circuit output	10,000 units	+
		Telecommunication services	Total volume of telecommunication services	10,000 yuan	+
	Industrial digital productivity	Network penetration rate	Number of Internet broadband access ports	No.	+
		Software services	Software business revenue	10,000 yuan	+
		Digital information	Optical cable line length per unit area	m/m ²	+
		E-commerce	E-commerce sales volume	10,000 yuan	+

Note: R&D = research and development; GDP = gross domestic product; SO₂ = sulfur dioxide. “+” denotes a positive indicator, for which a higher value represents a higher level of regional new quality productive forces; “—” denotes a negative indicator, for which a lower value represents a higher level of regional new quality productive forces.

4.3.2 Antecedent conditions

The antecedent conditions were organized into the three dimensions established in the theoretical framework: marketization level, marketization scale, and marketization environment. Each dimension was represented by two conditions.

Antecedent condition 1: Number of big data exchanges. This condition measures the formal institutional infrastructure available for data circulation and trading. Data exchanges provide organized venues for matching supply and demand, registering data products, standardizing transaction procedures, and establishing rules concerning pricing, compliance, and permitted use. For each province and year, the variable was measured as the cumulative number of officially established data exchanges in operation by the end of that year. Information was manually verified

using provincial government documents, development plans, official announcements, and exchange websites. A larger number indicates a more extensive formal infrastructure for data transactions, although it does not necessarily imply equivalent transaction volumes or operational capacities across exchanges.

Antecedent condition 2: Government data openness. Government data openness reflects the institutional and technical capacity of provincial governments to make public data accessible and reusable. The measure covers four dimensions: organizational readiness, platform capacity, data availability and quality, and data use. Annual provincial scores were obtained from the corresponding editions of the China Local Government Data Openness Report and its Open Data Forest Index [35]. For each province–year observation, the score reported for the corresponding year was used. This annual matching procedure avoided treating the value reported in a single edition as representative of the entire 2018–2023 period.

Antecedent condition 3: Informatization level. Regional informatization was measured using three components: human capital, industrial scale, and technological application. The corresponding indicators were employment in the information service industry, the output value of the information service industry, and software product revenue. Together, these indicators capture the availability of specialized personnel, the economic scale of information-related industries, and the commercialization of digital technologies. The indicators were obtained from the statistical yearbooks and Wind Economic Database [30–33] described in Section 4.2 and were combined using the entropy-weighting method. A common weighting structure derived from the pooled province–year sample was used to maintain comparability across years.

Antecedent condition 4: Share of e-commerce enterprises. E-commerce enterprises are major generators and users of transaction, consumer, logistics, and platform data. Their presence therefore indicates the extent to which data-driven commercial activities are embedded in the regional economy. The variable was calculated as the number of registered e-commerce-related enterprises divided by the total number of registered enterprises in the same province and year. E-commerce-related enterprises were identified using a consistent industrial-classification and business-scope screening rule across all regions and years. The resulting ratio was expressed as a percentage.

Antecedent condition 5: Marketization environment index. The selection of the five institutional dimensions followed the framework of the National Economic Research Institute (NERI) Marketization Index [36]. These dimensions cover the relationship between government and the market, the development of the non-state-owned economy, product-market development, factor-market development, and the development of market intermediaries and the legal environment. Annual provincial indicators for these dimensions were obtained from the statistical sources [30–33] and combined using the entropy-weighting method. Higher values indicate a more mature and predictable institutional environment for market exchange.

Antecedent condition 6: Government policy support. Government policy support reflects the attention provincial authorities devote to the development and regulation of data-factor markets. Provincial government work reports were collected for each region and year and processed using a consistent text-mining procedure. The coding dictionary covered terms relating to data elements, data transactions, public-data use, data-property rights, data security, and data-market development. The annual frequency of matched terms in each provincial government work report was used to represent the intensity of policy attention. This measure captures policy emphasis rather than policy implementation or expenditure [37].

Definitions, calculation procedures, units, and sources for the six antecedent conditions are summarized in Table 2.

4.4 Variable Calibration and Descriptive Statistics

Before the necessity and sufficiency analyses, the original variables were transformed into fuzzy-set membership scores ranging from 0 to 1. These scores represent the degree to which each province–year observation belongs to a specified set, such as the set of regions with high new quality productive forces or strong government data openness. Scores approaching 1 indicate strong membership, scores approaching 0 indicate strong non-membership, and a score of 0.5 represents maximum ambiguity regarding set membership.

The direct calibration procedure was applied. Because theoretically established external thresholds were unavailable for several conditions and the provincial observations differed substantially in economic scale, digital development, industrial structure, and institutional capacity, the 75th, 50th, and 25th percentiles of the pooled sample were used as the anchors for full membership, the crossover point, and full non-membership, respectively [27, 38]. The same calibration rule was applied to the outcome and all antecedent conditions. For negatively oriented indicators, the direction of calibration was reversed before fuzzy-set membership scores were calculated.

The percentile anchors define membership relative to the distribution of the 31 Chinese provincial-level regions during 2018–2023. The calibrated scores should therefore not be interpreted as universal thresholds applicable to other countries or observation periods. Observations calibrated to exactly 0.5 were increased to 0.501 to prevent their exclusion from truth-table minimization [38]. Table 3 reports the calibration anchors and descriptive statistics for the outcome and antecedent conditions.

Table 2. Measurement indicators of market-based allocation of data elements

Market-Based Allocation of Data Elements	First-Level Indicator	Indicator Description
Marketization level	Number of big data exchanges Government data openness level	Number of big data exchanges in each province Measured by the Open Data Forest Index
Marketization scale	Informatization scale Share of e-commerce enterprises	Human capital (employment in the information service industry) Industrial scale (output value of the information service industry) Technology application (software product revenue) Ratio of the number of e-commerce-related enterprises to the total number of enterprises in the region
Marketization environment	Marketization environment index Government policy support	Relationship between government and market Development of the non-state-owned economy Degree of product market development Degree of factor market development Development of market intermediary organizations and the legal-institutional environment Word frequency of data-element-marketization-related terms in government work reports

Table 3. Fuzzy-set calibration and descriptive statistics of variables

Condition	Fuzzy-Set Calibration			Descriptive Statistics			
	Full Membership	Crossover Point	Full Non-Membership	Maximum	Minimum	Median	SD
New quality productive forces	0.312	0.274	0.230	0.9	0.2	0.274	0.139
Number of big data exchanges	2	1	0	7	0	1	1.229
Government data openness level	34.899	6.665	0	78.56	0	6.665	22.523
E-commerce enterprise share	11.45	9.55	7.253	25.961	4.114	9.55	3.527
Informatization level	0.139	0.065	0.026	0.748	0.001	0.065	0.152
Marketization environment index	1.089	0.934	0.745	1.511	0.150	0.934	0.263
Government policy support	25	19	13	53	3	19	9.191

Note: SD = standard deviation. Full membership, crossover, and full non-membership correspond to the 75th, 50th, and 25th percentiles, respectively. Membership scores of exactly 0.5 are adjusted to 0.501.

5 Variable Analysis

5.1 Necessity Analysis of Individual Conditions

Before the sufficiency analysis, each antecedent condition and its negation were examined for necessity in relation to both high regional new quality productive forces and their absence. In set-theoretic terms, a condition is necessary when the outcome set is consistently a subset of the condition set. A consistency score of 0.90 was used as the principal threshold for identifying a potentially necessary condition [39]. Coverage was also reported to assess the empirical relevance of each necessity relationship, but it was not treated as an independent threshold that could compensate for consistency below 0.90.

For the panel analysis, pooled consistency summarizes the condition–outcome relationship across all 186

province–year observations. Between-group consistency was calculated separately for each year across provinces and was used to examine temporal variation. Within-group consistency was calculated for each province across the six observation years and was used to identify province-specific variation. Adjusted consistency distances were used as diagnostic measures of the extent to which the pooled relationship concealed temporal or provincial heterogeneity [29]. In accordance with the decision rule adopted in the dynamic analysis, an adjusted distance greater than 0.20 was treated as an indication that the corresponding relationship required disaggregated examination.

As reported in Table 4, none of the antecedent conditions or their negations reached a pooled consistency score of 0.90 for either high or low regional new quality productive forces. No individual condition was therefore identified as necessary. This result is consistent with the configurational premise of the study: high regional new quality productive forces arise from combinations of conditions related to marketization level, marketization scale, and marketization environment rather than from any isolated factor.

Table 4. Necessity condition analysis

Condition Variable	High Regional New Quality Productive Forces (Y)				Low Regional New Quality Productive Forces (Y)			
	Pooled Consistency	Pooled Coverage	Between-Group Consistency Adjustment Distance	Within-Group Consistency Adjustment Distance	Pooled Consistency	Pooled Coverage	Between-Group Consistency Adjustment Distance	Within-Group Consistency Adjustment Distance
Number of big data exchanges	0.725	0.693	0.195	0.362	0.464	0.431	0.264	0.544
~Number of big data exchanges	0.405	0.437	0.384	0.633	0.670	0.703	0.195	0.424
Government data openness level	0.649	0.668	0.125	0.367	0.429	0.430	0.330	0.580
~Government data openness level	0.447	0.446	0.163	0.578	0.669	0.649	0.173	0.378
Share of e-commerce enterprises	0.697	0.695	0.181	0.576	0.403	0.390	0.258	0.693
~Share of e-commerce enterprises	0.389	0.401	0.338	0.684	0.686	0.687	0.171	0.434
Informatization level	0.848	0.852	0.120	0.261	0.322	0.315	0.546	0.683
~Informatization level	0.318	0.325	0.375	0.700	0.848	0.845	0.233	0.180
Marketization environment index	0.835	0.840	0.134	0.463	0.300	0.293	0.340	0.721
~Marketization environment index	0.297	0.304	0.471	0.727	0.836	0.831	0.148	0.281
Government policy support	0.626	0.634	0.181	0.348	0.473	0.465	0.251	0.528
~Government policy support	0.471	0.479	0.240	0.517	0.627	0.620	0.183	0.415

Note: “~” denotes logical “not”.

Informatization and the marketization environment produced comparatively high pooled consistency scores of 0.848 and 0.835, respectively, with coverage values above 0.80. These conditions were frequently present among regions exhibiting high new quality productive forces, but neither reached the consistency threshold required for necessity. Their within-group adjusted consistency distances were 0.261 and 0.463, respectively, indicating that the corresponding relationships varied across provincial cases. The larger distance associated with the marketization environment points to particularly pronounced regional heterogeneity. The roles of informatization and market institutions should therefore be interpreted in relation to provincial economic structures, digital capacities, and stages of institutional development rather than as universally necessary conditions.

To further examine the between-group differences in these necessity relationships, Table 5 presents the year-by-year between-group consistency and coverage results for the six identified cases.

Table 5. Between-group difference analysis of necessary conditions

Case	Causal Combination	Indicator	Year					
			2018	2019	2020	2021	2022	2023
Case 1	~Number of big data exchanges–Regional new quality productive forces	Between-group consistency	0.557	0.546	0.482	0.299	0.277	0.274
		Between-group coverage	0.463	0.458	0.472	0.361	0.365	0.473
Case 2	~Share of e-commerce enterprises–Regional new quality productive forces	Between-group consistency	0.423	0.378	0.283	0.255	0.402	0.576
		Between-group coverage	0.404	0.390	0.368	0.333	0.382	0.482
Case 3	~Informatization level–Regional new quality productive forces	Between-group consistency	0.335	0.221	0.157	0.402	0.394	0.396
		Between-group coverage	0.322	0.297	0.269	0.329	0.328	0.369
Case 4	~Marketization environment index–Regional new quality productive forces	Between-group consistency	0.273	0.491	0.386	0.287	0.209	0.147
		Between-group coverage	0.314	0.377	0.341	0.278	0.239	0.217
Case 5	~Informatization level–~Regional new quality productive forces	Between-group consistency	0.907	0.686	0.577	0.963	0.972	0.987
		Between-group coverage	0.882	0.915	0.939	0.818	0.824	0.767
Case 6	Marketization environment index–~Regional new quality productive forces	Between-group consistency	0.412	0.184	0.213	0.282	0.347	0.365
		Between-group coverage	0.364	0.264	0.248	0.291	0.310	0.263

Note: “~” denotes logical “not”; condition variables with between-group consistency adjustment distance >0.2.

The X–Y scatter plots in Figure 2 provide a further examination of the necessity relationship identified in Case 5. The case distribution indicates that the observed consistency was not produced solely by observations with uniformly low membership in the relevant sets. The relationship between the absence of informatization and the absence of high new quality productive forces therefore had empirical relevance. However, because its consistency did not remain above 0.90 throughout the observation period, it cannot be classified as a stable necessary relationship for the complete panel.

The temporal pattern shown in Figure 3 further demonstrates this instability. Between-group consistency declined in 2019 and 2020 before increasing from 2021 onward. Although the decline coincided with the disruption associated with the COVID-19 pandemic, the current analysis does not establish a direct causal relationship. Changes in regional economic activity, digital investment, enterprise operations, policy implementation, or data availability may also have contributed to the observed movement. The later increase nevertheless indicates that the association between limited informatization and the absence of high new quality productive forces became more consistent during the final years of the study.

Overall, the necessity analysis produced two main findings. First, no single feature of market-oriented data allocation was indispensable for achieving high regional new quality productive forces. Second, informatization and the marketization environment were closely associated with the outcome, but their relevance varied across regions and periods. These results justify proceeding to the sufficiency analysis to identify the alternative combinations of conditions associated with the outcome.

5.2 Sufficiency Analysis of Condition Configurations

A sufficiency analysis was conducted using dynamic QCA to identify the combinations of conditions associated with high regional new quality productive forces. The truth table was constructed using a consistency threshold of 0.80, a proportional reduction in inconsistency (PRI) threshold of 0.75, and a case-frequency threshold of 1. No directional expectations were specified in advance for the antecedent conditions. The intermediate and parsimonious solutions were compared to distinguish core from peripheral conditions. Conditions appearing in both solutions

were treated as core conditions, whereas those appearing only in the intermediate solution were treated as peripheral conditions.

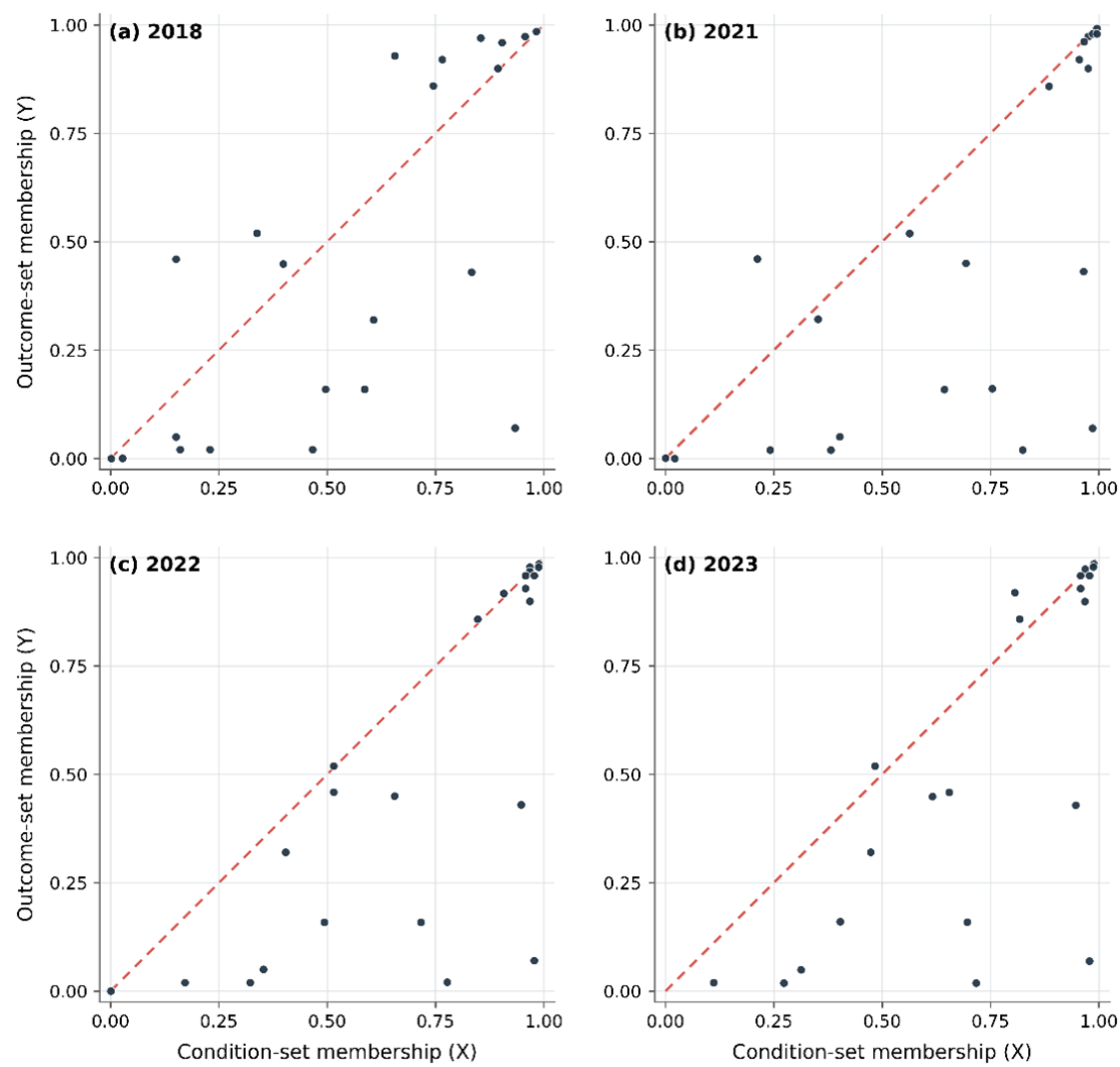


Figure 2. Necessary condition test scatter plots for Case 5: (a) 2018; (b) 2021; (c) 2022; (d) 2023

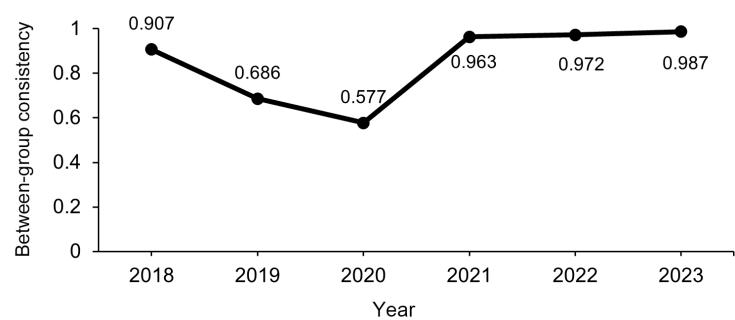


Figure 3. Changes in between-group consistency for Case 5

Table 6 reports six configurations associated with high regional new quality productive forces. Their consistency values ranged from 0.907 to 0.980, and their PRI values ranged from 0.858 to 0.974. All six pathways therefore exceeded the adopted consistency and PRI thresholds. Overall solution consistency reached 0.929, indicating that provincial cases represented by these configurations generally displayed high membership in the outcome set. Overall PRI was 0.911, suggesting limited simultaneous association with the negation of the outcome. Overall solution

coverage was 0.792, showing that the six configurations jointly accounted for a substantial proportion of the observed cases of high regional new quality productive forces.

Table 6. Configuration analysis results for driving regional new quality productive forces

Condition Variable	High Regional New Quality Productive Forces					
	S1a	S1b	S2a	S2b	S3a	S3b
Number of big data exchanges	⊗			•	•	
Government data openness level			•		⊗	•
Share of e-commerce enterprises	•	•			•	•
Informatization level			•	•		•
Marketization environment index	•	•	•	•		
Government policy		⊗				⊗
Consistency	0.924	0.958	0.972	0.980	0.907	0.972
PRI	0.872	0.945	0.964	0.974	0.858	0.963
Coverage	0.246	0.259	0.510	0.580	0.225	0.215
Unique coverage	0.008	0.006	0.034	0.027	0.016	0.006
Between-group consistency adjustment distance	0.018	0.015	0.045	0.030	0.076	0.048
Within-group consistency adjustment distance	0.110	0.145	0.221	0.303	0.239	0.075
Overall consistency				0.929		
Overall PRI				0.911		
Overall coverage				0.792		

Note: S1a–S3b denote the six solution configurations; “a” and “b” distinguish alternative configurations within the same solution group; PRI = proportional reduction in inconsistency; • indicates the presence of a core condition; ⊗ indicates the absence of a core condition; a blank indicates that the condition may be either present or absent.

The between-group consistency adjustment distances ranged from 0.015 to 0.076, all below 0.20. This result indicates that the six configurations remained broadly stable across the observation years. The within-group adjustment distances were higher for S2a, S2b, and S3a, reaching 0.221, 0.303, and 0.239, respectively. These values point to some variation among provincial cases within the same configuration and are examined further in the subsequent within-configuration analysis.

The six configurations demonstrate equifinality. High regional new quality productive forces were not associated with a single universal combination of conditions. Instead, different arrangements of data-market infrastructure, government data openness, informatization, e-commerce participation, market institutions, and policy support were capable of producing similarly high outcomes. On the basis of their core conditions, the configurations were classified into three broader models: the Scale–Environment Model, the Level–Scale–Environment Model, and the Level–Scale Model.

5.2.1 Scale–Environment Model

The Scale–Environment Model comprises configurations S1a and S1b. In both configurations, the share of e-commerce enterprises and the marketization environment index are core conditions. Their joint presence suggests that extensive commercial application of data, when supported by mature market institutions, can provide a viable pathway to high regional new quality productive forces. The difference between the two configurations lies in the absent condition: S1a includes the absence of big data exchanges, whereas S1b includes the absence of government policy support. The two pathways show how enterprise participation and the broader market environment may compensate for limitations in formal data-market infrastructure or direct policy intervention.

Configuration S1a combines a high share of e-commerce enterprises and a favorable marketization environment with the absence of well-developed big data exchanges. It indicates that regions with limited formal data-trading infrastructure may still support the circulation and productive use of data when commercial demand is sufficiently strong and the broader market environment allows resources to move across firms and sectors. E-commerce enterprises provide recurring settings for generating, processing, and applying data, while established market institutions support transactions, investment, and the combination of data with capital, technology, and labor.

Inner Mongolia, Gansu, and Ningxia are representative cases. These regions remain at a relatively early stage in the construction of formal data-trading infrastructure. Nevertheless, e-commerce has developed around locally distinctive industries, including livestock products in Inner Mongolia, traditional Chinese medicinal materials in Gansu, and goji berries in Ningxia. These activities have created practical settings for data generation and commercial

use. Improvements in the business environment and regional market mechanisms have also allowed enterprise activity to partly compensate for the limited presence of big data exchanges. S1a therefore represents an endogenous pathway in which sector-specific e-commerce and market institutions support data application before formal exchange infrastructure is fully established.

Configuration S1b combines a high share of e-commerce enterprises and a favorable marketization environment with the absence of strong government policy support. This pathway indicates that direct policy intervention is not indispensable when market demand, enterprise participation, and general institutional conditions are sufficiently developed. Active e-commerce ecosystems generate repeated demand for data-based applications, while an established market environment supports resource exchange, business formation, and innovation. Together, these conditions create a degree of market self-organization that can partly offset delayed or limited policy support.

Henan, Shaanxi, and Jiangxi illustrate this configuration. In these regions, some data-related and industrial policies have developed relatively slowly, and certain supporting arrangements remain incomplete. At the same time, the integration of e-commerce with agriculture, manufacturing, logistics, and locally distinctive industries has created considerable market activity. These regional e-commerce ecosystems have supported the circulation and commercial use of data even where direct policy effects have not been fully realized. S1b thus represents a market-led pathway in which enterprise participation and general institutional quality carry greater weight than direct government support.

5.2.2 Level–Scale–Environment Model

The Level–Scale–Environment Model includes configurations S2a and S2b. Both configurations contain informatization and the marketization environment as core conditions. Informatization supplies the technical capacity and human resources required to process and apply data, while the marketization environment provides the institutional setting for participation, exchange, and innovation. The difference lies in the condition through which data become accessible: government data openness is central to S2a, whereas big data exchanges are central to S2b. The model therefore identifies two alternative routes through which data access, digital capacity, and market institutions can operate jointly.

Configuration S2a combines high government data openness, a high level of informatization, and a favorable marketization environment. Government data openness increases the availability of public-sector information and creates opportunities for its reuse in commercial, technological, and governance applications. Informatization supplies the infrastructure and technical capacity required to process and integrate these resources. The marketization environment provides the institutional conditions under which firms and other organizations can convert accessible data into innovation and productive activity. This pathway is therefore characterized by coordination among public-data provision, regional digital capacity, and market institutions.

Tianjin, Zhejiang, and Chongqing are representative cases. These regions have developed public-data platforms alongside relatively strong informatization infrastructure and supportive market environments. Their experience indicates that public-data openness is most productive when it is accompanied by the capacity to process data and an institutional environment that encourages enterprise participation. The three conditions jointly connect government data resources with industrial transformation, urban digital services, and wider regional development.

Configuration S2b combines big data exchanges, informatization, and a favorable marketization environment. In this pathway, formal data-trading platforms provide institutional infrastructure for compliant circulation, transaction matching, and the specification of data products. Informatization supports data processing, integration, and application, while the marketization environment encourages participation by firms and other economic actors. The contribution of big data exchanges therefore depends on complementary technical and institutional capacity rather than on the existence of trading platforms alone.

Guangdong and Anhui illustrate this configuration. Both regions have developed data-trading platforms while investing in informatization infrastructure and improving their broader market environments. These conditions have supported the circulation of data resources and their integration with industrial and technological applications. S2b accordingly represents a platform-based pathway in which formal data exchanges, digital capacity, and market institutions jointly support the systematic development of new quality productive forces.

5.2.3 Level–Scale Model

The Level–Scale Model comprises configurations S3a and S3b. Both configurations contain a high share of e-commerce enterprises as a core condition, confirming the importance of enterprise participation and commercially grounded data-application settings. S3a combines big data exchanges with the absence of government data openness. S3b combines government data openness and informatization with the absence of government policy support. These configurations indicate that strong interaction between data access or transaction infrastructure and enterprise-level application can be associated with high regional new quality productive forces even when a particular public-sector condition is weak.

Configuration S3a combines the presence of big data exchanges and a high share of e-commerce enterprises with limited government data openness. This pathway indicates that formal market transactions and active commercial

demand can support data circulation even when public data are not extensively available. Big data exchanges provide channels for compliant transactions and resource matching, while e-commerce enterprises generate demand for data products and data-based services. Their interaction allows commercial and industrial data to be circulated and applied across production and service settings.

Jiangsu and Fujian are representative cases. Despite limitations in government data openness, both regions possess active digital industries, extensive e-commerce applications, and institutional arrangements for data circulation. The Suzhou Big Data Exchange, the Digital Fujian initiative, and regional industrial e-commerce clusters have provided transaction channels and application settings for data use. These mechanisms have partly offset weaknesses in public-data openness and supported the conversion of commercial and industrial data into productive capacity.

Configuration S3b combines a high share of e-commerce enterprises, government data openness, and informatization with the absence of strong direct policy support. The configuration suggests that an open-data environment and developed digital capacity can allow enterprises to obtain, process, and apply data from multiple sources even when additional policy intervention is limited. E-commerce activity expands the range of commercial applications, government data openness supplies reusable public information, and informatization provides the technical foundation for integration and analysis.

S3b should be interpreted as a cross-regional, application-led pathway rather than as a configuration represented exclusively by particular provinces or macro-regions. Its defining feature is the joint presence of government data openness, informatization, and e-commerce participation under conditions of limited direct policy support. These conditions allow existing public-data resources, digital infrastructure, and enterprise capabilities to support data application without continuous policy intervention. The regional analysis presented in Section 5.5 further examines whether this pathway is represented more strongly in particular parts of China.

Taken together, the three models show that market-oriented data allocation operates through different combinations of institutional conditions, technical capacity, and application settings. The Scale–Environment Model relies primarily on enterprise participation and the wider market environment. The Level–Scale–Environment Model combines data access, digital capacity, and supportive market institutions. The Level–Scale Model depends more directly on the interaction between data-market infrastructure or government data openness and enterprise-level application. These differences explain why a uniform policy model may be unsuitable for regions with different market structures, digital foundations, and governance capacities. They also provide a configurational basis for differentiated digital-governance and urban–regional development strategies.

5.3 Robustness Tests

The stability of the configuration results was examined by changing the raw-consistency and PRI thresholds and repeating the truth-table analysis. The first test increased the raw-consistency threshold from 0.80 to 0.85. This more restrictive specification produced six configurations identical to those obtained in the baseline analysis. The number of pathways, their core conditions, and their classification into the three broader models remained unchanged. The result indicates that the baseline configurations were not dependent on the original consistency threshold of 0.80.

The second test reduced the PRI threshold from 0.75 to 0.70. Under this specification, five configurations were obtained. S2a no longer appeared, whereas the other five configurations corresponded to those identified in the baseline analysis. The disappearance of S2a indicates that this particular pathway was more sensitive to the PRI specification than the remaining configurations and should therefore be interpreted with some caution.

Despite this variation, the central structure of the results remained largely unchanged. The Scale–Environment Model continued to be organized around the share of e-commerce enterprises and the marketization environment index. The remaining pathway in the Level–Scale–Environment Model continued to identify informatization and the marketization environment as core conditions. The Level–Scale Model continued to depend on a high share of e-commerce enterprises combined with either data-exchange infrastructure or government data openness and informatization.

The robustness tests therefore provide strong support for five of the six baseline configurations and qualified support for the overall three-model classification. Raising the consistency threshold did not alter any configuration, whereas changing the PRI threshold affected only S2a. The principal conclusion—that high regional new quality productive forces can arise through alternative combinations of marketization level, marketization scale, and marketization environment—remained stable. At the same time, the sensitivity of S2a should be acknowledged when interpreting the pathway based on government data openness, informatization, and the marketization environment.

5.4 Between-Configuration Analysis

The temporal stability of the six configurations was examined using between-group consistency and the corresponding adjustment distances. As shown in Figure 4, the annual between-group consistency values for all six configurations remained between 0.85 and 1.00, while all between-group adjustment distances were below 0.20. The results indicate that none of the configurations displayed substantial temporal heterogeneity during the observation

period. Each pathway maintained a generally consistent relationship with high regional new quality productive forces from 2018 to 2023.

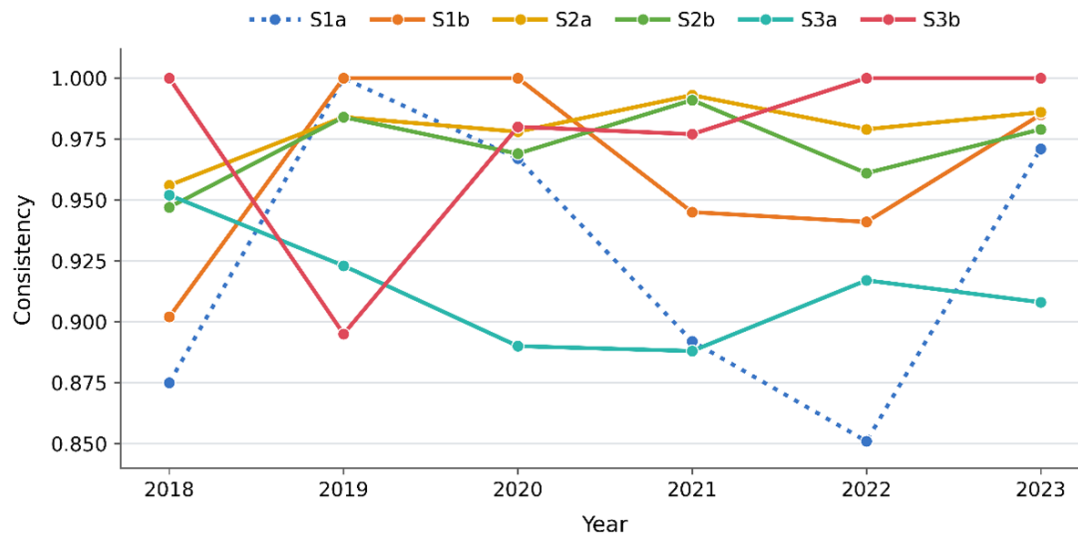


Figure 4. Annual between-group consistency of the six configurations, 2018–2023

Note: S1a–S3b denote the six solution configurations; “a” and “b” distinguish alternative configurations within the same solution group.

The absence of adjustment distances above 0.20 does not mean that the configurations remained completely unchanged. Their annual consistency values followed different trajectories. The Scale–Environment Model displayed a generally upward pattern, whereas the consistency of the Level–Scale Model gradually declined. The Level–Scale–Environment Model followed a V-shaped trajectory, with an initial decline followed by recovery. These movements do not overturn the overall stability of the configurations, but they reveal differences in how the pathways evolved during the six-year period.

The upward pattern of the Scale–Environment Model indicates that the combination of enterprise participation and a favorable marketization environment became increasingly consistent with high regional new quality productive forces. As data applications expanded, the wider institutional environment appears to have become more important in allowing e-commerce activity and other market-based uses of data to be translated into productive outcomes. The result is consistent with the view that commercial application requires predictable market rules, functioning intermediary institutions, and opportunities for resource exchange.

By contrast, the gradual decline of the Level–Scale Model suggests that reliance on data-access infrastructure and enterprise application without equally strong institutional-environment conditions may have become less consistently associated with the outcome. Big data exchanges, government data openness, informatization, and e-commerce activity remain important, but their presence alone may not be sufficient to sustain the same explanatory strength as regional data markets mature. This pattern should be interpreted as a change in relative consistency rather than evidence that the underlying conditions ceased to matter.

The V-shaped trajectory of the Level–Scale–Environment Model indicates greater short-term fluctuation in pathways that depend simultaneously on data access, digital capacity, and market institutions. The initial decline coincided with a period of wider economic and organizational disruption, after which consistency recovered. Because the analysis did not directly test the sources of this movement, it cannot be attributed to a particular external shock with certainty. The subsequent recovery nevertheless suggests that pathways combining technical, market, and institutional conditions retained their relevance over the longer observation period.

Overall, the between-configuration analysis shows that all three models remained temporally viable, although their relative consistency changed. The growing consistency of the Scale–Environment Model points to the increasing importance of enterprise participation and general market institutions. The declining tendency of the Level–Scale Model suggests that infrastructure and application capacity may require stronger institutional support as data markets develop. The recovery of the Level–Scale–Environment Model indicates that configurations based on multiple complementary conditions may experience short-term fluctuation while retaining longer-term explanatory relevance. These findings support differentiated policy design: urban and regional authorities should coordinate data infrastructure, enterprise application, and institutional development rather than relying on any single component of data marketization.

5.5 Within-Configuration Comparative Analysis

Within-configuration analysis was conducted to determine whether the consistency of a given pathway varied substantially across provincial cases. Unlike between-configuration analysis, which examines temporal stability, within-configuration analysis focuses on case-level heterogeneity within the same configurational pathway. A within-group consistency adjustment distance above 0.20 indicates that the pooled configuration may conceal meaningful differences among cases. Although S2a, S2b, and S3a were the configurations flagged by the adjusted-distance diagnostic, the regional comparison was reported for all six configurations to provide a consistent assessment and to identify geographical differences that might not be captured by the adjusted-distance measure alone.

As reported in Table 6, the within-group adjusted consistency distances for configurations S2a, S2b, and S3a were 0.221, 0.303, and 0.239, respectively. All three values exceeded the diagnostic threshold of 0.20, indicating that the consistency of these configurations varied across provincial cases. This variation may be associated with differences in regional economic structure, digital infrastructure, industrial specialization, institutional maturity, and the capacity to mobilize data resources.

To examine whether this case-level heterogeneity followed a systematic geographical pattern, a province-level within-group coverage score was calculated for each configuration. For province i and configuration k , the score was calculated as:

$$\text{Coverage}_{ik} = \frac{\sum_{t=2018}^{2023} \min(X_{ikt}, Y_{it})}{\sum_{t=2018}^{2023} Y_{it}}, \downarrow$$

where, X_{ikt} denotes the fuzzy-set membership of province i in configuration k in year t , and Y_{it} denotes the membership of province i in the outcome set of high regional new quality productive forces in the same year. The numerator measures the intersection between the configuration and the outcome across the six observation years, while the denominator represents the province's total membership in the outcome set over the same period. The calculation produced one within-group coverage score for each province under each configuration and therefore yielded 31 province-level observations for every configuration.

The 31 provincial-level regions were subsequently divided into eastern, central, and western groups following the regional classification adopted by Zhang [40]. Kruskal–Wallis tests were conducted separately for the six configurations to determine whether the distributions of their province-level within-group coverage scores differed among the three regional groups. The Kruskal–Wallis test was used because the province-level coverage scores were bounded between 0 and 1 and their distributions could not be assumed to satisfy the normality and homoscedasticity requirements of one-way analysis of variance.

The results are presented in Table 7. Regional differences were not statistically significant for S1a, S1b, S2a, S2b, or S3a. Although their mean province-level coverage scores varied among the eastern, central, and western regions, the Kruskal–Wallis tests did not reject the null hypothesis that the scores followed the same distribution across the three regional groups. These results indicate that no systematic regional difference was detected for these five configurations.

Table 7. Kruskal–Wallis tests for regional differences in province-level within-group coverage

Configuration	Sample	East	Central	West	Mean	SD	H	p -Value
S1a	31	0.166	0.276	0.380	0.277	0.279	3.546	0.169
S1b	31	0.230	0.203	0.269	0.238	0.207	0.192	0.908
S2a	31	0.584	0.542	0.396	0.500	0.332	2.503	0.286
S2b	31	0.553	0.460	0.367	0.456	0.310	2.421	0.298
S3a	31	0.256	0.115	0.428	0.286	0.265	5.341	0.069
S3b	31	0.227	0.191	0.556	0.345	0.340	7.087	0.028

Note: S1a–S3b denote the six solution configurations; “a” and “b” distinguish alternative configurations within the same solution group. SD = standard deviation. For each configuration, one within-group coverage score was calculated for each province using its six annual observations from 2018 to 2023. East, Central, and West report the mean province-level coverage scores for the corresponding regional groups. A separate Kruskal–Wallis test was conducted for each configuration to examine whether the distributions of these scores differed among the three groups.

S3b was the only configuration for which a statistically significant regional difference was detected ($H = 7.087$, $p = 0.028$). Its mean province-level within-group coverage was higher in western China than in the eastern and central regions. This result indicates stronger representation of S3b among western provincial cases during the study period. It does not mean that the configuration was exclusive to western China, because provinces belonging to the pathway may also be found in other regional groups.

The conditions defining S3b provide a possible interpretation of this regional difference. The configuration combines government data openness, informatization, and a high share of e-commerce enterprises with the absence of

strong direct policy support. The higher mean coverage observed in western China suggests that some western provinces may have developed combinations of public-data access, digital processing capacity, and enterprise participation that support data application without intensive direct policy intervention. This interpretation concerns relative configuration coverage and does not imply that western China as a whole possesses stronger digital infrastructure or higher new quality productive forces than the eastern and central regions.

S3b therefore represents a cross-regional application-led pathway with comparatively stronger western representation. Its regional distribution also illustrates the heterogeneity within China's broad geographical divisions: some western provinces may possess relatively developed open-data platforms, informatization capacity, and e-commerce activity even though the western region is often described as having weaker average digital and market foundations.

The descriptive results further show that the average coverage of the six configurations was generally higher in central and western China than in eastern China. One possible explanation is that recent digitalization and green-transformation initiatives in central and western regions have generated more concentrated development patterns. Provinces at similar stages of digital and industrial development may also be represented by the same configurations more consistently. These interpretations remain tentative, however, because the Kruskal–Wallis tests did not identify statistically significant regional differences for five of the six pathways.

The lower average coverage observed in eastern China should not be interpreted as evidence of weaker new quality productive forces. Eastern provinces generally possess more diversified industrial systems, more mature digital markets, and a wider range of development trajectories. This diversity may reduce the proportion of cases represented by any single configuration. In other words, lower configuration coverage may reflect greater pathway differentiation rather than inferior development performance. The eastern region may rely on more varied combinations of data infrastructure, enterprise capabilities, market institutions, and policy arrangements that are not fully represented by one dominant model.

Overall, no statistically significant regional difference was detected for five of the six configurations, whereas S3b showed a significant difference in province-level within-group coverage across the three regional groups. S3b was represented more strongly in western China, but it remained a cross-regional rather than an exclusively western pathway. The result reinforces the need for place-sensitive digital-governance and urban-regional development policies. Regions should not simply replicate a nationally uniform data-market model; policy design should reflect the particular combination of public-data access, digital capacity, enterprise participation, and institutional conditions already present in each location.

6 Conclusions and Implications

6.1 Research Conclusions

Drawing on transaction cost theory, resource orchestration theory, and institutional theory, this study examined how market-oriented data allocation was associated with regional new quality productive forces across 31 provincial-level regions in China from 2018 to 2023. Dynamic QCA was used to identify alternative sufficient pathways and to evaluate their temporal stability and regional variation. The analysis produced four main conclusions.

First, no individual antecedent condition constituted a necessary condition for high regional new quality productive forces. Although informatization and the marketization environment displayed comparatively high necessity consistency and coverage, neither reached the required consistency threshold of 0.90. Regional new quality productive forces therefore cannot be attributed to a single data exchange, policy intervention, technological condition, or institutional factor. They arise from the joint operation of several conditions whose roles vary across configurations.

Second, six sufficient configurations were identified. Their individual consistency values ranged from 0.907 to 0.980, while overall solution consistency and coverage reached 0.929 and 0.792, respectively. The configurations were classified into three broader models: the Scale–Environment Model, the Level–Scale–Environment Model, and the Level–Scale Model. The Scale–Environment Model was organized around e-commerce participation and the marketization environment. The Level–Scale–Environment Model combined informatization and market institutions with either government data openness or big data exchanges. The Level–Scale Model depended on enterprise participation together with data-exchange infrastructure or the combination of government data openness and informatization. These models demonstrate equifinality: regions with different digital, market, and institutional foundations can reach high levels of new quality productive forces through different pathways.

Third, the six configurations were generally stable over time. Their between-group consistency values remained between 0.85 and 1.00, and all between-group adjustment distances were below 0.20. The pathways nevertheless followed different temporal trajectories. The consistency of the Scale–Environment Model generally increased, whereas that of the Level–Scale Model gradually declined. The Level–Scale–Environment Model followed a V-shaped pattern. These movements indicate changes in the relative explanatory consistency of the pathways rather than the emergence or disappearance of the configurations. The results suggest that enterprise participation and general market institutions became more closely associated with the outcome over the study period, while pathways relying primarily on data access and application capacity displayed a weaker tendency in later years.

Fourth, no statistically significant regional difference was detected for S1a, S1b, S2a, S2b, or S3a. S3b was the only configuration for which province-level within-group coverage differed significantly among eastern, central, and western China. Its mean coverage was higher in western China, indicating that the pathway was more strongly represented among western provincial cases during the study period. S3b combined government data openness, informatization, and a high share of e-commerce enterprises with the absence of strong direct policy support. The result suggests that some western provinces developed an application-led pathway based on public-data access, digital capacity, and enterprise participation. However, S3b was not exclusive to western China, and the regional difference should not be interpreted as evidence that western regions had higher overall levels of new quality productive forces.

Overall, market-oriented data allocation contributes to regional new quality productive forces through combinations of transactional infrastructure, resource-application capacity, and institutional conditions. The results do not support a uniform development model. Effective policy requires alignment between existing regional capabilities and the conditions that are absent or underdeveloped. This configurational perspective is particularly relevant to urban and regional development because provincial data institutions shape the digital infrastructure, enterprise ecosystems, public-data access, and innovation resources available to the cities located within them.

6.2 Policy Implications

Develop differentiated systems for government data openness and institutional support. Government data openness should be treated as part of a wider institutional and technical system rather than as a stand-alone publication activity. Provincial and urban authorities should improve data standards, metadata quality, updating procedures, interoperability, access conditions, and mechanisms for lawful reuse. Central and western regions with less developed public-data systems should give priority to basic platform capacity and the usability of released datasets. Legal rules, intermediary services, transaction procedures, fair-competition mechanisms, and data-security responsibilities also require attention because access alone does not ensure productive application.

Eastern regions generally possess more mature digital infrastructure and open-data systems. Their policy focus can move toward public-data authorization, clearer allocation of data-related rights, cross-jurisdictional circulation, and the development of replicable governance practices. Cooperation among eastern, central, and western regions should not be limited to transferring technical platforms. It should also cover data standards, institutional experience, professional training, and mechanisms for connecting regional data resources with urban public services and industrial applications.

Use e-commerce ecosystems as practical settings for data application. The Scale–Environment Model shows that e-commerce participation and a favorable market environment can partly compensate for limitations in formal data exchanges or direct policy support. Central and western regions can build on locally distinctive industries to develop e-commerce clusters and digital supply chains. Agriculture, specialized manufacturing, cultural products, logistics, and regional services can provide concrete settings in which data are generated, circulated, and applied. Such ecosystems can support early-stage data marketization before formal data exchanges become fully developed.

This finding does not imply that e-commerce platforms can permanently replace data exchanges. Their functions differ. E-commerce ecosystems create application demand and commercial data flows, whereas exchanges provide standardized and compliant channels for formal data transactions. As regional data markets mature, the two should become more closely connected. Eastern regions can take a leading role in linking e-commerce platforms, data exchanges, and digital-industry clusters. Platform firms can gradually develop services related to data processing, transaction support, compliance management, and ecosystem coordination while participating in the design of transparent pricing and transaction arrangements.

Coordinate data infrastructure, informatization, and market institutions. The Level–Scale–Environment Model indicates that data exchanges and government data openness are most effective when they are supported by informatization and a functioning market environment. Investment in platforms alone is unlikely to produce lasting results when regions lack digital professionals, data-processing capacity, enterprise demand, or predictable institutional rules. Data-market policies should therefore be coordinated with broadband and computing infrastructure, digital-skills development, software and information-service industries, and support for enterprise-level applications.

Central and western regions should continue to address gaps in information infrastructure and digital human capital, but investment decisions should be linked to identifiable industrial and public-service needs. Government intervention is most useful when it establishes common standards, removes institutional barriers, and enables participation by firms and other organizations. It should not substitute for enterprise initiative or create platforms with limited transaction and application demand.

Eastern regions face a different challenge. Their diversified industrial systems and mature digital markets make reliance on a single pathway less appropriate. Policy should permit experimentation across data trading, public-data reuse, industrial digitalization, urban governance, and cross-regional data circulation. Greater attention should also be given to interoperability among existing platforms and to the institutional coordination needed to prevent fragmentation.

Adopt place-sensitive strategies for urban and regional development. Provincial data-market policies have direct consequences for cities because urban areas concentrate enterprises, infrastructure, public services, and data-intensive activities. Provincial authorities should allow cities with different industrial structures and digital capacities to follow different development pathways while maintaining common rules for data security, interoperability, and lawful circulation. Major metropolitan areas may focus on cross-sectoral data integration and advanced data services, whereas smaller cities may begin with public-data applications and the digital transformation of locally important industries.

The regional concentration of S3b also shows that western regions should not be treated as uniformly dependent on direct policy support. Where government data openness, informatization, and enterprise participation are already relatively strong, policy can focus on removing remaining transaction barriers and allowing existing market actors to expand data applications. Direct intervention should be calibrated to local conditions rather than applied with the same intensity across all regions.

6.3 Limitations and Future Research

Several limitations should be considered when interpreting the findings. First, the spatial unit of analysis was the provincial-level region. This scale allowed consistent comparison across China and captured broad institutional and economic differences, but it may conceal substantial variation among cities within the same province. Provincial capitals, metropolitan regions, resource-based cities, and smaller prefecture-level cities may differ considerably in digital infrastructure, government data openness, industrial composition, and enterprise capabilities. The results should therefore be interpreted as provincial configurational patterns rather than direct evidence of city-level mechanisms.

Future research could extend the analysis to prefecture-level cities or metropolitan regions. City-level data would make it possible to test whether the provincial pathways remain valid at a finer spatial scale and to identify the conditions under which provincial policies are translated into urban development outcomes. Multilevel designs could also examine how provincial institutions interact with city-level infrastructure, industrial specialization, and governance capacity.

Second, the analysis covered the period from 2018 to 2023. Although the panel structure made it possible to examine temporal stability, the six-year period was relatively short compared with the institutional development cycle of data-factor markets. The observation window did not capture the complete transition from the initial establishment of data exchanges and open-data platforms to the formation of mature transaction and application systems. Some of the pathways identified here may change as legal frameworks, digital infrastructure, and enterprise practices continue to develop.

Longer panel datasets should be used when additional annual observations become available. An extended observation period would allow future studies to distinguish short-term fluctuation from structural change, identify possible transitions between configurations, and assess whether the relative importance of market institutions, government data openness, and enterprise participation changes as data markets mature.

Third, government policy support was measured through the frequency of data-market-related terms in provincial government work reports. This measure captures the degree of policy attention but cannot fully distinguish among regulatory, financial, infrastructural, and organizational policy instruments. It also does not directly measure implementation quality, expenditure, enforcement, or actual policy outcomes. Regions with similar term frequencies may therefore differ considerably in the substance and effectiveness of their policies.

Future studies could apply more detailed policy-text coding to distinguish policy objectives, instruments, target groups, and implementation mechanisms. Text-based indicators could also be combined with policy expenditure, implementation records, regulatory documents, project completion data, or interviews with public officials and market participants. Such evidence would provide a more precise account of how different forms of government intervention interact with market and technological conditions.

Fourth, the temporal comparability of government data openness requires further attention. The Open Data Forest Index captures several relevant dimensions of provincial data openness, but the availability and construction of annual observations may not correspond perfectly to the complete 2018–2023 study period. Any inconsistency in annual coverage could reduce the precision of the dynamic comparison. Future research should use fully harmonized annual indicators or reconstruct comparable measures directly from provincial open-data platforms.

Finally, QCA identifies set-theoretic associations and alternative sufficient pathways, but it does not by itself establish the detailed causal processes through which each configuration produces the outcome. The regional cases discussed in this study illustrate the substantive meaning of the pathways but do not replace in-depth case evidence. Comparative case studies, interviews, process tracing, and mixed-method designs could be used to examine how data exchanges, public-data platforms, enterprises, and government institutions interact within particular regions. Combining these approaches with dynamic QCA would provide a fuller understanding of how market-oriented data allocation contributes to new quality productive forces and how these mechanisms influence urban and regional development.

Author Contributions

Conceptualization, C.W. and M.Y.L.; methodology, C.W.; software, M.Y.L.; validation, M.Y.L.; data curation, M.Y.L.; writing—original draft preparation, M.Y.L.; writing—review and editing, C.W.; supervision, C.W. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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